

ATTENTION BASED LANE SEGMENTATION AND MULTI-CLASS TRAFFIC SIGN DETECTION USING LIGHT WEIGHT DEEP LEARNING MODELS

A PROJECT REPORT

Submitted by

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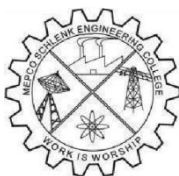
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BONAFIDE CERTIFICATE

Certified that this project report titled “**ATTENTION BASED LANE SEGMENTATION AND MULTI-CLASS TRAFFIC SIGN DETECTION USING LIGHT WEIGHT DEEP LEARNING MODELS**” is the bonafide work of **KIRAN KUMAR V** (Reg. No. 9517202203082), **MITHUNRAJ R** (Reg. No. 9517202203095) who carried out the research under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Vehicle transportation has become an essential part of human transport systems; however, it is responsible for numerous deaths, injuries, and financial losses. The issue of road safety has received considerable attention recently, which helped create advanced driver assistance systems (ADAS). These devices enhance human capabilities of recognizing, analyzing, and responding to diverse conditions with the help of intelligent technologies. Among the various ADAS perception applications, there are two main areas that have a considerable impact on the decision-making process while driving.

Lane detection is essential for detecting drivable lanes, proper road alignment, and lane departure warnings in case the vehicle moves outside the safe area. Traffic sign detection assists in the recognition of semantic concepts like speed limits, stop signs, and pedestrian crossings, which are critical for informed decision-making. However, though individually lane detection and traffic sign detection are crucial for self-driving cars, most research papers discuss both lane and traffic sign detection separately, thus rendering the perception module inadequate and not functional in practice.

The present study proposes a novel framework for lane detection and traffic sign detection that combines the two processes efficiently. Lane detection employs an encoder-decoder architecture similar to LLDNet, but enhanced with the CBAM module to help the network focus on lane features while excluding road cracks, shadows, and object occlusions from its vision. The dataset used for training and testing the lane detector included 14,999 images of road scenes. On the other hand, the traffic sign detection model relies on YOLOv8 technology and uses a custom dataset comprising three types of traffic signs – child pedestrian crossing, speed limit signs, and stop signs.

From experiments, one can notice that our lane detection model that utilizes both LLDNet and CBAM models achieves high accuracy (98.73%), precision (97.39%), recall (97.00%), Intersection-over-union (IoU) (94.54%), and Dice Score (97.19%). All of these measures have shown to be superior to the other segmentation models, requiring only 0.26 million parameters during training. In addition to the first branch that detects lanes, the second branch, which helps us detect traffic signs, uses YOLOv8 model and achieves precision (96.6%), recall (97.1%), mean Average Precision at IOU (98.9%) and mean Average Precision at IOU (50 to 95 as 86.8%).

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADAS	Advanced Driver Assistance Systems
BCE	Binary Cross-Entropy
BN	Batch Normalization
CBAM	Convolutional Block Attention Module
CNN	Convolutional Neural Network
C2f	Cross Stage Partial with Two Convolutions (Fast)
DFL	Distribution Focal Loss
FCN	Fully Convolutional Network
FPN	Feature Pyramid Network
FPS	Frames Per Second
GTSDb	German Traffic Sign Detection Benchmark
GTSRB	German Traffic Sign Recognition Benchmark
IoU	Intersection over Union
LLDNet	Lightweight Lane Detection Network
mAP	Mean Average Precision
NMS	Non-Maximum Suppression
PAN	Path Aggregation Network
ReLU	Rectified Linear Unit
RGB	Red Green Blue (colour space)
ROI	Region of Interest
SSD	Single Shot MultiBox Detector
SPPF	Spatial Pyramid Pooling - Fast
SVM	Support Vector Machine
TTS	Text-to-Speech
YOLO	You Only Look Once

SDGs ALINGMENT

SDG Icons	Relevance of SDG in the context of project	Section Name(s)	Page number(s)
 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	Lightweight deep learning-based ADAS system using LLDNet+CBAM and YOLOv8 demonstrates innovative AI-based infrastructure for real-time lane detection and traffic sign recognition, enabling scalable, efficient, and intelligent transportation systems for smart mobility applications.	System Architecture (Lane Detection & Traffic Sign Detection), Software Implementation, Results and Discussion	13–20, 22–26 & 33–41
 11 SUSTAINABLE CITIES AND COMMUNITIES	Real-time lane detection and traffic sign recognition improve road safety, reduce accidents, and enhance driver awareness in urban environments, supporting safer transportation systems and sustainable smart city mobility solutions.	Introduction, Problem Statement, Results and Discussion	1–2 & 33–41

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The means of transport being used across the globe in moving people and goods from one point to another is road transport. According to international organizations, more than 1.35 million people lose their lives annually due to road accidents. Road accidents rank as one of the main causes of deaths in the world. For developing countries like India, which face issues of increasing urbanization, growing number of vehicles, and bad roads, the situation becomes much worse. Apart from huge casualties, road accidents lead to economic losses due to medical bills, automobile repair costs, and non-productive working hours. This has motivated researchers to come up with intelligent transport systems to limit dependence on the driver's decision-making ability.

Among the best technologies adopted to solve the issue are the Advanced Driver Assistance Systems (ADAS). ADAS encompasses various computer and electronic systems that detect and evaluate the environment surrounding the vehicle and notify in case of any risks. Modern ADAS technology includes a wide range of elements including forward collision warnings, automatic emergency braking, lane departure alerts, adaptive cruise control, lane departure assist, traffic sign recognition, and many others. This technology is instrumental in ensuring that there are minimal accidents.

In comparison with all other components of ADAS systems, lane and traffic sign detection are considered to be the simplest ones because of their visual recognition. Lane detection implies that the driver is informed about the accuracy of driving between the lane lines. Thus, one cannot think about changing lanes or automatic steering in absence of such a system. Traffic signs give an opportunity to realize the situations when certain rules should be followed. The speed limits inform the driver about the maximum speed allowed in this particular place. The stop sign means that the car should immediately halt.

Thirdly, recent developments in deep learning revolution have also influenced the discussed tasks. Firstly, CNN is an effective tool for hierarchical feature extraction without using fragile pipeline-based architectures. Encoder-decoder architectures (e.g., U-Net, SegNet) made it possible to conduct pixel-to-pixel segmentation of road scenes. Attention mechanisms (e.g., Convolutional Block Attention Module (CBAM)) allow to draw attention

to some parts of the input image or feature map. Consequently, such network architecture provides robust performance under challenging conditions. Meanwhile, with the help of YOLO and its modifications, object detection has become quick and accurate at once. The ability to perform object detection on several objects in a single run of the neural network is an invaluable property for ADAS systems.

However, despite numerous achievements in machine vision, very few papers have been dedicated to detecting lanes and traffic signs as mutually dependent tasks. In reality, both aspects should go hand-in-hand. Indeed, the vehicle should drive within the appropriate lane and recognize traffic signs.

1.2 Problem Statement

The problem addressed in this research is the development of a robust perception algorithm that facilitates the simultaneous detection of lanes and traffic signs based on images and videos in the real world. The objective of the framework being developed addresses the limitations associated with the existing frameworks, which include the following:

- The techniques adopted in conventional frameworks to detect lanes using algorithms such as Hough transform, edge detection, and geometric fitting are ineffective in complicated scenarios such as curved roads, blurred lanes, nighttime driving, and poor weather conditions because of pre-programmed algorithms.
- The deep learning algorithms used to detect lanes have high accuracy but require a large number of parameters and computing power, which makes them computationally expensive for implementation in low-computing vehicles.
- The existing frameworks consider traffic signs to be independent of lanes, thereby limiting their awareness capabilities.
- The existing frameworks adopt separate pipelines for detecting traffic signs and lanes, which restricts the use of their similarities.

1.3 Objectives

The primary objectives of this project are as follows:

- 1.Design an advanced lightweight encoder-decoder network that integrates CBAM in its design to achieve excellent binary lane segmentation from road-scene images.
- 2.Implement a real-time multiclass traffic sign detector that can recognize and categorize child-pedestrian crossing signs, speed limit signs, and stop signs using YOLOv8.
- 3.Integrate both lane detection and traffic sign detection branches into one pipeline, targeting ADAS-inference outputs with dual modalities of visual and audio alerts.
- 4.Analyze the efficiency of the suggested framework using conventional evaluation measures like accuracy, precision, recall, F1-score, IoU, Dice, mAP@50, and mAP@50-95.
- 5.Determine the significance of different components used in our approach through a comprehensive ablation study involving the implementation of the CBAM block, selection of appropriate loss functions, data pre-processing methods, and post-processing methods.
- 6.Demonstrate the capability of the suggested model to produce consistent outcomes in different environmental settings, such as nighttime, rainy weather, and daylight.

1.4 Scope of the Work

The scope of the study will be limited to the identification of road scenes using visual information gathered from a single front camera. LiDAR, radar, and multiple camera systems will not be used in this study. In the case of the lane detector component, binary classification will be applied to distinguish between lane and non-lane pixel categories. The traffic signs detector component will identify three specific types of traffic signs, which include the child pedestrian crossing, speed limit, and stop signs. While the design of the system architecture targets real-time operations, the current implementation of the system targets offline applications.

1.5 Organisation of the Report

This document is organized as follows. First, in Chapter 2, a comprehensive literature review regarding lane detection and traffic sign detection approaches through conventional methods and deep learning methods is offered. Second, the proposed method's architecture is presented in Chapter 3. Third, the software implementation process is detailed in Chapter 4. Fourth, the evaluation measures for the experiment are introduced in Chapter 5. Fifth, the results of the experiment are shared in Chapter 6. Sixth, the ablation study is outlined in Chapter 7. Seventh, conclusions are made in Chapter 8.

CHAPTER 2

LITERATURE SURVEY

In the past several decades, many scientific studies have been conducted in relation to the analysis of road scenes for improving driving safety and driver independence. The development of lane detection and traffic sign detection technology has been divided into two generations; the first one is based on feature extraction and heuristics, while the second generation applies deep learning. Both generations have achieved remarkable results, had their own strengths and weaknesses, and raised problems to be researched.

2.1 Traditional Methods for Lane Detection

The very first lane detection systems relied on basic image processing techniques based on the characteristics of lanes. Lanes usually have a bright white or yellow color, are shaped in lines, and exhibit clear contrasts in comparison with the dark-colored ground surface. Hence, applying such edge detectors as the Canny algorithm to detect edges in the image and then using the Hough Transform to fit the edge points into a line was a feasible idea. As proved by Voisin et al. (2005), this technique along with a Kalman filter could be implemented in real time in the case of structured roads. Nevertheless, lane markings had to be straight in order to use the Hough Transform.

A number of methods have been proposed to solve modeling issues in lane detection tasks. For example, one can refer to Aly's (2008) research in which the author proposes a practical approach towards lane detection based on IPM transformation. IPM transformation was applied for eliminating perspective differences between road images obtained from the front camera perspective and from the top perspective of the road. The practicality of the suggested approach was confirmed with the help of using Gaussian filtering and RANSAC technique applied for calculating polynomial models of roads.

Wang et al. (2004) introduced the B-Snake model, an active contour approach where flexible snake curves were initialised near lane boundaries and allowed to deform iteratively until they converged to the lane edges. Unlike rigid line fitting, the B-Snake could accommodate curved and irregular lane shapes. However, the snake initialisation required good starting estimates, and the iterative optimisation was computationally expensive for real-time use. Kang and Jung (2003) proposed dynamic programming for road lane segmentation, formulating it as an energy minimisation problem under geometric

constraints. This approach produced smooth, consistent lane estimates but was sensitive to the choice of energy function parameters.

Several researchers also introduced preprocessing and filtering strategies to enhance lane visibility before detection. Hu et al. (2020) proposed a Dynamic Region of Interest (DROI) method that adaptively searched for lane regions using interpolated control points, improving computational efficiency by limiting processing to relevant image areas. Gao et al. (2017) applied Gabor filtering as a preprocessing step to enhance linear lane-like structures before edge extraction. Fan et al. (2020) addressed illumination problems by combining top-hat morphological transformation, Hough fitting, and Kalman filtering for robust lane tracking. Wei and others experimented with different edge operators, such as the Robert operator, to reduce noise sensitivity. Andrei et al. (2021) applied probabilistic Hough transformation with optimised region selection to improve runtime performance.

More specialised geometric representations were explored to handle complex road geometries. Wennan et al. (2006) used hyperbola-based models for lane boundaries in complex intersections. Wang and Shen (1998) applied Catmull-Rom splines for fitting lane structures in noisy images. Li et al. (2014) developed a parallel-snake approach that simultaneously tracked both lane boundaries, enforcing the constraint that they remain approximately parallel. Yoo et al. (2017) proposed a vanishing point estimation method based on probabilistic voting, which improved robustness in challenging scenes by leveraging the geometric constraint that lane lines converge toward a single vanishing point.

Despite decades of development, traditional lane detection methods share a fundamental weakness: they depend on manually designed features and geometric assumptions that are inherently brittle. A system designed for clear highway markings may fail entirely on rural roads with no painted lines. A method tuned for daylight may produce noisy outputs under artificial lighting. These limitations drove the eventual shift to data-driven deep learning approaches.

2.2 Traditional Methods for Traffic Sign Detection

Traditional traffic sign detection was built on the observation that traffic signs are designed to be visually distinctive. Most signs have standardised shapes (circles, triangles, rectangles, hexagons) and highly saturated colours (red, blue, yellow, white) that stand out from natural backgrounds. Early detection pipelines exploited these properties through colour-based segmentation: image pixels were transformed into a colour space such as HSV

or normalised RGB, and regions matching the target sign colours were extracted as candidates. Shape verification was then applied, using contour analysis or template matching, to confirm that candidate regions had the correct geometric form.

These methods worked adequately in controlled laboratory conditions but deteriorated rapidly in the field. Illumination changes — from bright sunshine to shadows to night driving with artificial light — dramatically altered the apparent colour of signs. Sign degradation, dirt, or partial occlusion removed salient features. Background clutter containing objects with similar colours or shapes produced false positives. Mogelmose et al. (2012) conducted an extensive review of classical traffic sign detection and concluded that handcrafted feature-based systems were not robust enough for deployment in real-world ADAS environments, particularly for applications requiring high recall.

2.3 Deep Learning Based Lane Detection Methods

The introduction of deep convolutional neural networks fundamentally changed the lane detection landscape. Unlike handcrafted pipelines, CNNs automatically learn hierarchical representations of visual features directly from data. At shallow layers, they detect basic structures like edges, corners, and oriented lines. At deeper layers, they learn more abstract representations such as lane continuity, road geometry, and the semantic distinction between the drivable lane region and its surroundings. This data-driven learning eliminates the need for manual feature engineering and enables generalisation across diverse road types, weather conditions, and camera setups.

Kim and Park (2017) were among the early pioneers of end-to-end deep learning for lane detection, formulating the problem as semantic segmentation using SegNet, an encoder-decoder architecture that produces a pixel-level class label map. They demonstrated that transfer learning from ImageNet could significantly improve performance on lane datasets with limited annotations, and that their model required no hand-tuned post-processing. This represented a major departure from classical methods.

Neven et al. (2018) addressed the multi-lane detection problem more explicitly by introducing LaneNet, which treated each lane as a separate instance rather than assigning a single binary label. LaneNet combined a segmentation branch with an embedding branch, where pixels belonging to the same lane were pulled closer in an embedding space while pixels from different lanes were pushed apart. This instance segmentation approach allowed

the system to naturally handle lane changes and multiple parallel lanes, which was difficult for binary segmentation methods.

Zou et al. (2020) observed that single-frame processing discards valuable temporal information available in video streams. They proposed a hybrid CNN-RNN framework that fed features from consecutive frames into a recurrent module, allowing the network to leverage temporal continuity for more stable and accurate lane detection. This temporal awareness was especially beneficial for handling occlusions and momentarily missing lane markings. Chen and Xiang (2022) similarly introduced a spatio-temporal attention module that used contextual information from preceding frames to improve detection under occluded lane markings.

Zhang et al. (2021) introduced RS-Lane, which incorporated split attention mechanisms to improve lane estimation in challenging traffic situations. The attention mechanism allowed the network to dynamically weight different feature channels based on their relevance, effectively suppressing noise from shadows, road cracks, and crossing vehicles. This was an important step toward more robust lane detection under real-world complexity.

Lightweight architectures for real-time embedded deployment attracted considerable attention. Hou et al. (2019) proposed a self-attention distillation approach for lightweight lane detection, training a small student network to mimic the attention maps of a larger teacher network. This knowledge distillation strategy allowed the compact model to achieve accuracy close to the teacher while reducing inference time significantly. Ko et al. (2021) developed PINet, a point-instance-based network that represented lanes as sequences of points rather than dense segmentation maps, substantially reducing computational requirements. Tabelini et al. (2021) proposed PolyLaneNet, which directly regressed polynomial coefficients representing lane boundaries, achieving very high FPS on the TuSimple benchmark. Tabelini et al. (2021) also proposed LaneATT, using anchor-based attention mechanisms for robust lane detection under occlusion.

Munir et al. (2021) introduced LDNet, which incorporated event-based vision inputs to improve robustness against illumination variation and motion blur, areas where traditional frame-based cameras struggle. Multi-task learning approaches also emerged, with Lee (2021) proposing a network based on MobileNetV2 that simultaneously estimated drivable areas and lane lines, improving efficiency for real-time autonomous driving. Haris et al.

(2022, 2023) contributed a series of works focusing on decoder enhancement, multi-scale convolution, and asymmetric kernel design for lane detection in complex environments, achieving strong performance with compact architectures.

Khan et al. (2022) specifically proposed LLDNet (Lightweight Lane Detection Network), an encoder-decoder architecture designed for computational efficiency without sacrificing accuracy. LLDNet used a compact encoder with four convolutional blocks and a lightweight decoder with transposed convolution layers for upsampling. Its relatively small parameter count made it attractive as a base architecture for embedded deployment. The proposed work in this project builds upon LLDNet and enhances it with CBAM attention to further improve its lane feature discrimination capability.

2.4 Deep Learning Based Traffic Sign Detection Methods

Traffic sign detection underwent a parallel revolution driven by deep learning. The introduction of large-scale benchmark datasets was a critical enabling factor. Stallkamp et al. (2011) presented the German Traffic Sign Recognition Benchmark (GTSRB), containing over 50,000 images across 43 traffic sign categories, and later (2012) organised the GTSRB competition that attracted participation from more than 25 teams. The winning entries, including multi-scale CNNs by Sermanet and LeCun, demonstrated that deep networks could surpass human-level performance on traffic sign recognition. This benchmark provided a standardised evaluation platform that accelerated research progress dramatically.

Houben et al. (2013) extended the benchmark to detection (rather than recognition) with the German Traffic Sign Detection Benchmark (GTSDB), which provided full road-scene images requiring localisation and classification of signs. This more challenging benchmark reflected the true difficulty of detecting signs in cluttered driving environments.

The emergence of general-purpose object detection frameworks brought powerful tools to traffic sign detection. Ren et al. (2017) proposed Faster R-CNN, a two-stage detector where a Region Proposal Network (RPN) first generated candidate sign locations, which were then classified by a CNN. Faster R-CNN achieved strong localisation accuracy on various benchmarks, including traffic signs. However, its two-stage design introduced latency that made it less suitable for real-time ADAS applications.

One-stage detectors addressed this latency problem. Liu et al. (2016) proposed SSD (Single Shot MultiBox Detector), which predicted bounding boxes and class probabilities in a single forward pass by applying detection heads at multiple feature map scales. This

multi-scale approach allowed SSD to detect signs at different distances and sizes efficiently. Redmon et al. (2016) introduced the landmark YOLO (You Only Look Once) framework, which divided the image into a grid and directly predicted bounding boxes and class probabilities for each cell in a single unified pass, achieving unprecedented speed. Subsequent versions (YOLOv2, YOLOv3, YOLOv4, YOLOv5) progressively improved accuracy while maintaining speed.

Jocher et al. (2023) released Ultralytics YOLOv8, the latest evolution of the YOLO family, incorporating several architectural improvements including a new C2f (Cross Stage Partial with Two Convolutions - Fast) backbone, an anchor-free detection head, and an improved loss function. YOLOv8 achieved state-of-the-art results on standard benchmarks while providing a user-friendly API for custom dataset training. Its combination of accuracy, speed, and ease of use makes it particularly attractive for ADAS traffic sign detection applications, as adopted in the proposed work.

2.5 Summary and Research Gap

The literature review reveals that both lane detection and traffic sign detection have matured significantly through the application of deep learning. However, the following gaps remain unaddressed:

- The majority of high-performance lane detection systems require large model sizes with millions of parameters, limiting their suitability for resource-constrained embedded automotive platforms.
- While attention mechanisms have shown promise for lane detection robustness, their integration into lightweight architectures specifically designed for ADAS deployment remains limited.
- Lane detection and traffic sign detection are almost universally studied as separate, independent problems in existing literature, despite the fact that practical ADAS requires both simultaneously.
- Unified frameworks that fuse structural road understanding (lane geometry) with semantic road understanding (traffic sign meaning) into a single coherent inference pipeline are rare and underexplored.

The proposed work directly addresses these gaps by developing a lightweight attention-guided lane detection model and integrating it with a real-time traffic sign detector within a single unified ADAS-oriented perception framework.

Table 2.1: Combined Comparison of Representative Lane Detection and Traffic Sign Detection Studies

Author	Year	Method	Task	Key Contribution	Limitation
Voisin et al.	2005	Hough + Kalman	Lane Detect.	Efficient lane tracking via line fitting + temporal filter	Fails on curved roads, clutter
Aly	2008	IPM + RANSAC	Lane Detect.	Bird's-eye view transformation for urban lanes	Sensitive to camera calibration
Kim & Park	2017	SegNet Transfer Learning	Lane Detect.	End-to-end lane segmentation without post-processing	Poor on unstructured roads
Neven et al.	2018	LaneNet (Instance Seg.)	Lane Detect.	Instance separation enables multi-lane handling	High computational complexity
Zou et al.	2020	CNN-RNN Hybrid	Lane Detect.	Temporal context improves lane continuity	Increased inference latency
Zhang et al.	2021	RS-Lane Split Attention	Lane Detect.	Attention-guided robustness in challenging scenes	Attention adds model complexity
Khan et al.	2022	LLDNet (Lightweight)	Lane Detect.	Compact encoder-decoder designed for embedded use	No attention mechanism
Stallkamp et al.	2011	GTSRB Benchmark	Sign Detect.	Standard benchmark for recognition tasks	Recognition only, no localisation
Houben et al.	2013	GTSDb Benchmark	Sign Detect.	Real-world detection benchmark with localisation	Limited sign category diversity
Redmon et al.	2016	YOLO (v1)	Sign Detect.	Real-time one-stage object detection	Struggles with small distant signs
Ren et al.	2017	Faster R-CNN	Sign Detect.	Strong localisation via region proposals	Too slow for real-time ADAS
Jocher et al.	2023	YOLOv8	Sign Detect.	State-of-the-art speed + accuracy with anchor-free head	Dataset quality dependency

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 Overview of Proposed Framework

The proposed system is an integrated vision-based perception framework designed for Advanced Driver Assistance Systems. The framework accepts a single road-scene image or video frame as input and processes it through two parallel deep learning branches: the Lane Detection Branch and the Traffic Sign Detection Branch. The outputs of both branches are subsequently fused in a post-processing module to generate a unified ADAS output comprising lane boundary visualisation, traffic sign bounding boxes and labels, and driver alert messages. The overall system architecture is illustrated in Figure 1.

The framework is designed around three guiding principles. First, computational efficiency: both branches are designed to be lightweight so that the complete system can operate at real-time speeds without requiring heavy computing resources. Second, robustness: attention mechanisms and strong data augmentation ensure that the system performs reliably across diverse road types, lighting conditions, and weather. Third, integration: rather than treating the two tasks independently, the proposed system fuses their outputs at the inference level to provide a holistic road-scene understanding that neither branch could achieve alone.

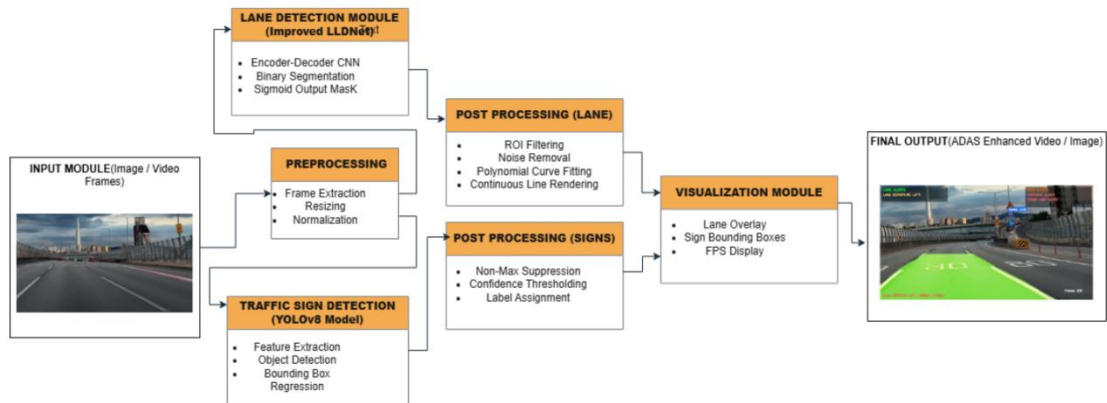


Fig. 1: Complete ADAS System Pipeline Block Diagram

3.2 Lane Detection Branch

The lane detection branch formulates lane perception as a binary semantic segmentation task. Every pixel in the input image is classified as either lane (class 1) or background (class 0). This pixel-level formulation is preferred over line-fitting or polynomial regression approaches because it naturally handles complex lane geometries

including curves, broken markings, road edges, and merged lanes, without requiring geometric assumptions.

3.2.1 Preprocessing

Raw road-scene images exhibit significant variation in resolution, aspect ratio, and pixel intensity range. Before feeding images into the neural network, two preprocessing steps are applied. First, the image is resized to a fixed resolution of 80×160 pixels (height×width). This resolution was selected after experimentation to balance spatial detail preservation with computational efficiency. Second, pixel intensities are normalised by dividing by 255, mapping values from the range [0, 255] to [0, 1]. Normalisation ensures that the network receives inputs with a stable distribution, accelerating training convergence and improving generalisation. These operations are mathematically expressed as:

$$Ir \in \mathbb{R}^{Hr \times Wr \times 3} \quad (\text{resized image, } Hr \times Wr) \quad (3.1)$$

$$In(x, y, c) = \frac{Ir(x, y, c)}{255} \quad (3.2)$$

3.2.2 Encoder

The encoder is a sequence of four convolutional blocks, each consisting of a Conv2D layer, Batch Normalisation, ReLU activation, and MaxPooling. The role of the encoder is to hierarchically transform the input image into compact, semantically rich feature representations by progressively reducing spatial resolution while increasing the number of feature channels.

At the first block (En-1), the encoder processes low-level visual features such as edges, intensity gradients, and colour boundaries. These features capture the local contrast between lane markings and the road surface. At En-2 and En-3, the receptive field expands, and the network learns mid-level features including lane shape, orientation, and road texture patterns. At the deepest block (En-4), the encoder captures high-level semantic information including the full spatial extent of the drivable lane, road curvature, and the relative positioning of lane boundaries within the scene. The fundamental convolution operation at each spatial location (i,j) is:

$$Z(i, j) = \sum_m \sum_n X(i + m, j + n) \cdot K(m, n) + b \quad (3.3)$$

where X is the input feature map, K is the learned convolution kernel of size m×n, b is a bias term, and Z(i,j) is the output response. The ReLU nonlinearity applied after each convolution is:

$$f(z) = \max(0, z) \quad (3.4)$$

This introduces nonlinearity into the network, enabling it to learn complex, non-linear feature relationships that are essential for understanding the variability of road scenes.

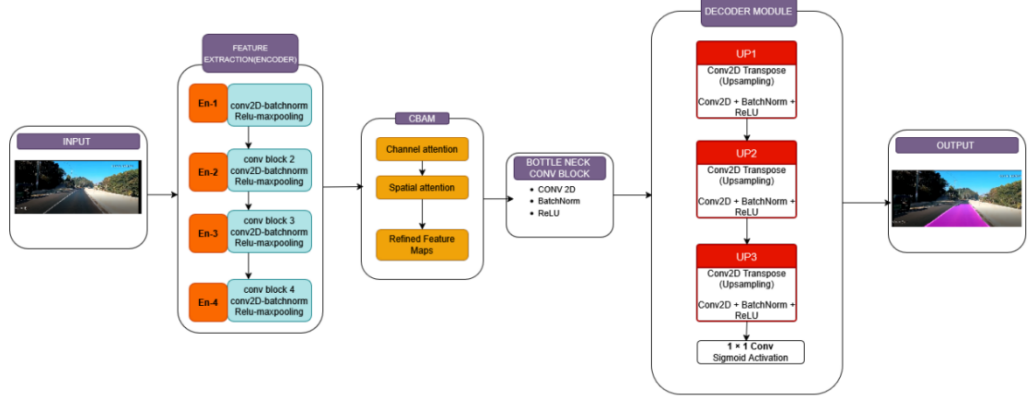


Fig. 2: Lane Detection Branch – Encoder-CBAM-Decoder Architecture

3.2.3 Convolutional Block Attention Module (CBAM)

The Convolutional Block Attention Module (CBAM) is a lightweight attention mechanism that refines feature representations by applying attention along both the channel dimension and the spatial dimension. This dual attention strategy allows the network to explicitly learn what features are relevant (channel attention) and where in the image to focus (spatial attention). The detailed architecture of CBAM is shown in Figure 3.

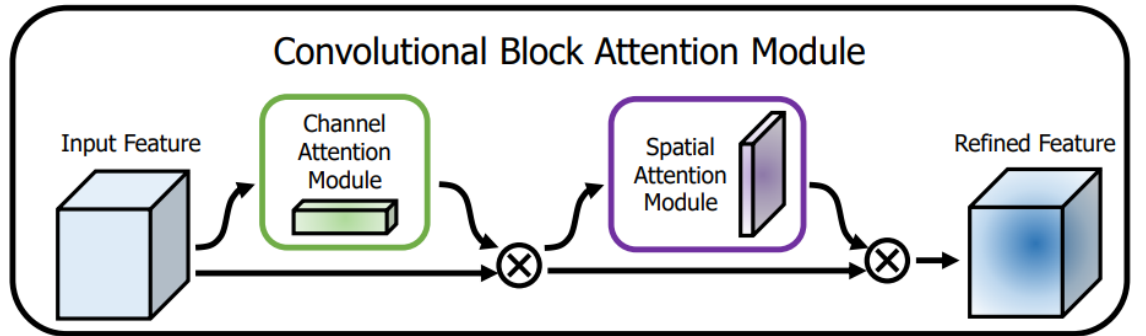


Fig. 3: CBAM Architecture – Channel Attention and Spatial Attention Modules

The channel attention module computes a channel descriptor by applying both global average pooling and global max pooling along the spatial dimensions of the feature map F of size $C \times H \times W$. Both descriptors capture different statistical properties of each channel: average pooling captures the mean response while max pooling captures the most

discriminative response. These two descriptors are independently passed through a shared two-layer Multi-Layer Perceptron (MLP) with a ReLU activation in between. The outputs are summed and passed through a sigmoid function to produce the channel attention weights M_c of size $C \times 1 \times 1$:

$$M_c(F) = \sigma \left(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F)) \right) \quad (3.5)$$

where σ denotes the sigmoid activation function. The channel-refined feature map F' is obtained by element-wise multiplication:

$$F' = M_c(F) \otimes F \quad (3.6)$$

The spatial attention module then processes F' to generate a spatial attention map that indicates which locations in the image are most informative. Global average pooling and global max pooling are applied along the channel dimension to produce two 2D descriptor maps of size $1 \times H \times W$. These are concatenated and passed through a single 7×7 convolutional layer followed by sigmoid activation to produce the spatial attention map M_s :

$$M_s(F') = \sigma(\text{Conv}_{7 \times 7}[\text{AvgPool}(F'); \text{MaxPool}(F')]) \quad (3.7)$$

The final CBAM output F_r is obtained by spatial-wise multiplication:

$$F_r = M_s(F') \otimes F' \quad (3.8)$$

By sequentially applying channel and spatial attention, CBAM effectively suppresses irrelevant background noise such as shadows, road cracks, vegetation, and vehicle bodies, allowing the network to concentrate its representational capacity on the actual lane structure. Empirical results in the ablation study confirm that CBAM contributes a 2.43% improvement in lane detection accuracy over the baseline without attention.

3.2.4 Decoder

The decoder progressively upsamples the refined feature map F_r back to the original image resolution, reconstructing a dense pixel-level segmentation output. The decoder consists of a bottleneck convolution block followed by three upsampling blocks (UP1, UP2, UP3). Each upsampling block uses a transposed convolution (ConvTranspose2D) to double the spatial resolution, followed by a Conv2D layer with Batch Normalisation and ReLU activation to refine the upsampled features. The final output is produced by a 1×1 convolution layer that maps the feature channels to a single probability map, followed by a sigmoid activation function:

$$\hat{Y}(x, y) = \sigma(z(x, y)) = \frac{1}{1 + e^{-z(x, y)}} \quad (3.9)$$

where $z(x, y)$ is the decoder output logit and $\hat{Y}(x, y) \in [0, 1]$ is the predicted probability that pixel (x, y) belongs to the lane class. The binary lane mask is obtained by thresholding:

$$\hat{M}_{\text{lane}}(x, y) = \begin{cases} 1 & \text{if } \hat{Y}(x, y) \geq \tau, \\ 0 & \text{else} \end{cases} \quad (3.10)$$

where $\tau = 0.5$ is the threshold value. Pixels with probability above τ are classified as lane, while those below are classified as background.

3.2.5 Output and Loss Function

The optimisation objective for training the lane detection branch is a combination of Binary Cross-Entropy (BCE) loss and Dice loss. BCE loss penalises incorrect pixel-wise predictions and is defined as:

$$L_{\text{BCE}} = -\frac{1}{N} \sum_i [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (3.11)$$

where N is the total number of pixels, y_i is the ground truth binary label, and $\hat{y}_i$ is the predicted probability. Dice loss directly optimises the overlap between predicted and ground-truth segmentation regions and is defined as:

$$L_{\text{Dice}} = 1 - \frac{2 \sum_i y_i \hat{y}_i}{\sum_i y_i + \sum_i \hat{y}_i} \quad (3.12)$$

The combined loss is:

$$L_{\text{total}} = L_{\text{BCE}} + L_{\text{Dice}} \quad (3.13)$$

The Dice loss component is particularly important because it explicitly optimises the IoU-related metric, compensating for class imbalance: in road scene images, background pixels vastly outnumber lane pixels, which would cause BCE alone to focus excessively on correctly classifying background. The ablation study confirms that the combined loss reduces validation loss from 0.1245 (BCE only) to 0.0654 (combined), a reduction of approximately 47%.

3.3 Traffic Sign Detection Branch

The traffic sign detection branch formulates the problem as a multi-class object detection task, where the system must simultaneously localise and classify multiple traffic signs in a road-scene image. Unlike lane detection, which assigns a label to every pixel, object detection requires predicting a bounding box, a class label, and a confidence score for each detected object. The branch is implemented using YOLOv8, the most recent

generation of the YOLO (You Only Look Once) family, selected for its superior balance between detection accuracy and inference speed.

3.3.1 YOLOv8 Architecture

YOLOv8 adopts a three-stage architecture comprising a backbone, a neck, and a detection head, as illustrated in Figure 4. Each stage plays a distinct and complementary role in the detection pipeline.

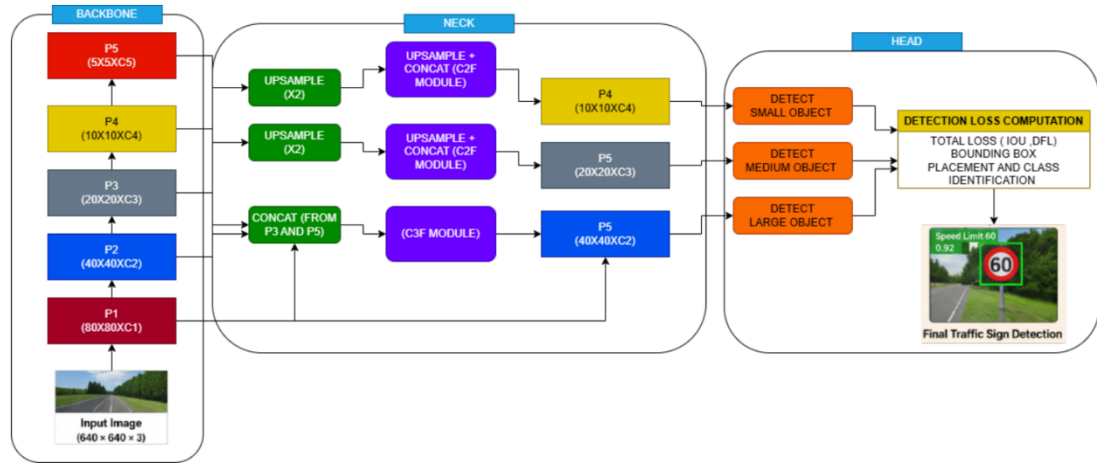


Fig. 4: YOLOv8 Architecture – Backbone, Neck (FPN+PAN) and Detection Head

The backbone is responsible for multi-scale feature extraction from the input image. It begins with a convolutional stem (CBS block: Conv2D + Batch Norm + SiLU activation) that reduces the spatial resolution while increasing the number of feature channels. This is followed by a series of C2f (Cross Stage Partial with Two Convolutions - Fast) blocks, which are modified Cross Stage Partial modules that improve gradient flow during backpropagation by splitting the feature map and processing parts of it through bottleneck layers. At the end of the backbone, a Spatial Pyramid Pooling - Fast (SPPF) module aggregates multi-scale contextual information by applying three consecutive max-pooling operations with a small kernel, effectively creating a multi-scale representation without expensive pyramid construction.

The neck consists of a Feature Pyramid Network (FPN) combined with a Path Aggregation Network (PAN). The FPN propagates semantic information from deep layers (which have large receptive fields) down to shallow layers (which have fine spatial resolution) through top-down upsampling and feature concatenation. The PAN then propagates localisation information from shallow layers up to deep layers through bottom-up path augmentation. This bidirectional feature fusion ensures that all detection scales receive both semantic context and spatial detail, which is critical for detecting traffic signs

of varying sizes — from large, close signs that occupy a significant portion of the image to small, distant signs that are only a few pixels wide.

The detection head in YOLOv8 is anchor-free, representing a major architectural departure from earlier YOLO versions. Instead of predicting offsets relative to predefined anchor boxes, the anchor-free head directly predicts the centre coordinates, width, and height of bounding boxes as continuous values. This eliminates the need for anchor hyperparameter tuning and reduces the complexity of the prediction task. Each detection level (P3, P4, P5 for three different scales) outputs a tensor predicting bounding box parameters, objectness scores, and class probabilities. The final set of detections is defined as:

$$D = \{(b_i, c_i, s_i)\}, \quad b_i = (x_i, y_i, w_i, h_i) \quad (3.14)$$

where b_i represents the bounding box centre coordinates and dimensions, c_i is the class label, $s_i = p_{\text{obj}} \cdot p_{\text{cls}}$ is the composite confidence score, and N_d is the number of surviving detections after post-processing. The Intersection over Union (IoU) between predicted and ground-truth boxes is:

$$\text{IoU}(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|} \quad (3.15)$$

The training loss for YOLOv8 combines bounding box regression loss, classification loss, and distribution focal loss (DFL):

$$L_{\text{sign}} = \lambda_{\text{box}} \cdot L_{\text{box}} + \lambda_{\text{cls}} \cdot L_{\text{cls}} + \lambda_{\text{dfl}} \cdot L_{\text{dfl}} \quad (3.16)$$

where L_{box} is the CIOU regression loss that accounts for bounding box overlap, centre distance, and aspect ratio simultaneously, providing more informative gradient signals than simple IoU loss.

3.3.2 Post-Processing

The raw output of the YOLOv8 detection head contains a large number of candidate bounding boxes, most of which correspond to background regions or redundant overlapping detections. Two post-processing steps are applied. First, low-confidence detections with $s_i < \tau_{\text{conf}} = 0.12$ are discarded. Second, Non-Maximum Suppression (NMS) with an IoU threshold of 0.50 eliminates redundant overlapping boxes by retaining only the highest-confidence box when multiple boxes overlap significantly. This yields a clean, final set of traffic sign detections ready for display and alert generation.

3.4 Integrated ADAS Fusion Module

The fusion module combines the outputs of both detection branches into a unified visual output and alert generation pipeline. The lane binary mask $\hat{M}_{\text{lane}}$ is overlaid on the original road-scene image as a semi-transparent coloured region (typically green), providing the driver with a clear visualisation of the safe drivable region. Simultaneously, each detected traffic sign from the detection set D is annotated on the image with a bounding box, class label, and confidence score.

Based on the detected sign classes, the fusion module generates context-aware driver alerts. If a stop sign is detected, an alert 'STOP SIGN DETECTED' is generated. If a speed limit sign is detected, the alert includes the detected sign class. If a child-pedestrian crossing sign is detected, a warning alert is issued. These alerts are communicated through a text-to-speech (TTS) engine, providing spoken warnings to the driver. A scheduling mechanism with controlled inter-alert delays prevents concurrent audio outputs. The mathematical representation of the fusion function is:

$$V_{\text{final}} = \Gamma(I, \hat{M}_{\text{lane}}, D), \quad A = \Phi(\hat{M}_{\text{lane}}, D) \quad (3.17)$$

where V_{final} is the final fused visualisation, $\Gamma(\cdot)$ is the fusion/overlay function, A is the alert set, and $\Phi(\cdot)$ is the alert generation function that maps lane context and detected sign classes to appropriate driver notifications.

CHAPTER 4

SOFTWARE IMPLEMENTATION

4.1 Feature Extraction Techniques

Feature extraction is the process of transforming raw input data into a compact, informative numerical representation that captures the characteristics most relevant to the task at hand. In deep learning, feature extraction occurs automatically through the learned convolutional filters of the network. However, understanding the underlying principles of these automated feature extraction processes is important for interpreting model behaviour and designing appropriate architectures.

4.1.1 Convolutional Feature Extraction

In the LLDNet+CBAM encoder, feature extraction begins with the application of small learnable convolutional filters (typically 3×3) that scan across the input image. Each filter responds to a different visual pattern. Early filters in the first convolutional block learn to detect simple features like horizontal and vertical edges, corners, and colour gradients. These correspond directly to the visual appearance of lane markings, which are characterised by strong horizontal intensity gradients at their boundaries.

As the feature maps pass through subsequent convolutional blocks, the effective receptive field grows (because each subsequent layer 'sees' a larger region of the original image). The filters begin to respond to more complex, composite structures: the full width of a lane marking, the shape of a complete lane boundary, the spatial relationship between two lane lines, and eventually the entire lane region within the road context. This hierarchical feature learning is what allows deep networks to understand lane geometry from raw pixels without any manually engineered feature descriptors.

4.1.2 Batch Normalisation

Each convolutional layer in the encoder and decoder is followed by Batch Normalisation (BN). BN normalises the activations within a mini-batch by subtracting the batch mean and dividing by the batch standard deviation, then applying learnable scale and shift parameters γ and β :

$$\text{BN}(x) = \gamma \cdot \frac{x - \mu_{\text{batch}}}{\sqrt{\sigma_{\text{batch}}^2 + \varepsilon}} + \beta \quad (4.1)$$

where μ_{batch} and σ^2_{batch} are the batch mean and variance, and ε is a small constant for numerical stability. BN serves two purposes: it reduces internal covariate shift,

stabilising and accelerating training convergence, and it acts as a mild regulariser, reducing the risk of overfitting on limited training data.

4.1.3 Multi-Scale Feature Representation in YOLOv8

For traffic sign detection, feature extraction operates at multiple spatial scales simultaneously. Traffic signs can appear at very different sizes depending on their distance from the camera. A stop sign 10 metres away may occupy a 100×100 pixel region in the image, while the same sign at 50 metres may occupy only 20×20 pixels. A single-scale feature extractor would either miss small distant signs or fail to precisely localise large close signs.

YOLOv8 addresses this through the SPPF module in the backbone and the FPN+PAN neck. The SPPF applies max-pooling operations with kernels of different effective sizes, generating feature responses at multiple scales. The FPN creates a top-down pathway that fuses high-level semantic features from the deepest backbone layer with fine-grained spatial features from shallower layers. The PAN then creates a bottom-up pathway to reinforce localisation information at all scales. The three detection heads (P3 at 1/8 scale, P4 at 1/16 scale, P5 at 1/32 scale) specialise in detecting objects of small, medium, and large sizes respectively.

4.2 Classification and Detection Algorithms

4.2.1 Binary Semantic Segmentation

Lane detection is implemented as binary semantic segmentation, which assigns exactly one of two class labels — lane (positive class) or background (negative class) — to every pixel in the image. This is fundamentally a per-pixel binary classification problem. The LLDNet+CBAM model produces a probability map $\hat{Y} \in [0,1]^{(H \times W)}$ where each value represents the network's confidence that the corresponding pixel belongs to the lane class. The binary mask is obtained by thresholding at $\tau = 0.5$.

The key metric for evaluating segmentation quality is the Intersection over Union (IoU), also known as the Jaccard Index, which measures the overlap between the predicted mask and the ground-truth mask:

$$\text{IoU} = \frac{|\text{Pred} \cap \text{GT}|}{|\text{Pred} \cup \text{GT}|} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \quad (4.2)$$

An IoU of 1.0 indicates perfect overlap, while 0.0 indicates no overlap. The Dice coefficient, closely related to IoU, is also commonly used:

$$\text{Dice} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}} = \frac{2 \cdot \text{IoU}}{1 + \text{IoU}} \quad (4.3)$$

4.2.2 Multi-Class Object Detection

Traffic sign detection is implemented as multi-class object detection, which requires predicting the bounding box coordinates, class label, and confidence score for each sign instance present in the image. The key distinction from segmentation is that the output is a set of object-level predictions rather than a pixel-level map.

YOLOv8 frames detection as a direct regression problem: given an input image, the network directly outputs all bounding boxes and class predictions in a single forward pass. The network is trained using a composite loss comprising CIoU bounding box regression loss, binary cross-entropy classification loss, and distribution focal loss (DFL). The DFL term encourages the network to predict sharp, confident bounding box edge probability distributions, improving localisation precision.

4.2.3 Data Augmentation

Both branches benefit from data augmentation, which artificially increases the effective size and diversity of the training dataset. For lane detection, augmentation includes random horizontal flipping, brightness and contrast adjustment, Gaussian noise addition, and random cropping. For traffic sign detection, YOLOv8's built-in augmentation pipeline includes Mosaic augmentation (combining four images into one), random scaling, translation, shear, HSV colour jitter, and horizontal flipping. These augmentations expose the models to a wider variety of lighting conditions, viewing angles, and image qualities, significantly improving generalisation to unseen test images.

4.3 Mathematical Formulations

The complete mathematical pipeline of the lane detection branch can be summarised as a sequence of transformations from the raw input image $I \in \mathbb{R}^{(H \times W \times 3)}$ to the binary lane mask $\hat{M}_{\text{lane}} \in \{0,1\}^{(H \times W)}$:

1. Preprocessing: $I \rightarrow I_n \in [0,1]^{(H_r \times W_r \times 3)}$ via resizing and channel-wise normalisation.
2. Encoding: $F_e = E(I_n; \theta_e)$ where $E(\cdot)$ is the encoder function with parameters θ_e .
3. Attention refinement: $F_r = \text{CBAM}(F_e) = M_s(M_c(F_e) \otimes F_e) \otimes (M_c(F_e) \otimes F_e)$.
4. Decoding: $F_d = D(F_r; \theta_d)$ where $D(\cdot)$ is the decoder function with parameters θ_d .
5. Probability map: $\hat{Y} = \sigma(F_d)$ via pixel-wise sigmoid activation.
6. Binary mask: $\hat{M}_{\text{lane}} = 1 \{ \hat{Y} \geq 0.5 \}$.

7. Loss: $L_{\text{total}} = L_{\text{BCE}}(\hat{Y}, Y) + L_{\text{Dice}}(\hat{Y}, Y)$ during training.

For the traffic sign detection branch, the key mathematical quantities are the composite confidence score $s_i = p_{\text{obj},i} \cdot p_{\text{cls},i}$, the IoU metric for localisation evaluation, and the mean Average Precision (mAP) for overall detection performance assessment. mAP@50 is the mean of per-class Average Precision values computed at an IoU threshold of 0.50, while mAP@50-95 averages AP over IoU thresholds from 0.50 to 0.95 in steps of 0.05, providing a more comprehensive evaluation of localisation quality.

$$\text{mAP@50} = \frac{1}{C} \sum_k \text{AP}_k^{0.50}, \text{ where } C = \text{number of classes} \quad (4.4)$$

$$\text{mAP@50-95} = \frac{1}{C} \sum_k \frac{1}{T} \sum_t \text{AP}_k^t, \quad t \in \{0.50, 0.55, \dots, 0.95\} \quad (4.5)$$

4.4 Software Environment

The complete software stack includes the following tools and libraries: Python 3.9+ as the primary programming language; TensorFlow 2.x with Keras API for the lane detection model; Ultralytics YOLOv8 (built on PyTorch) for the traffic sign detector; OpenCV for image and video processing, frame capture, and visualisation; NumPy and SciPy for numerical computations; Matplotlib and Seaborn for result visualisation and analysis; Pandas and Scikit-learn for data management and metric computation.

Table 4.1 Experimental Setup and System Configuration

Category	Lane Detection Branch	Traffic Sign Detection Branch
Task	Binary semantic segmentation	Multi-class object detection
Model	LLDNet-inspired encoder-decoder + CBAM	YOLOv8 (anchor-free one-stage detector)
Framework	TensorFlow 2.x / Keras	Ultralytics YOLOv8 (PyTorch)
Dataset	Custom lane dataset	Custom traffic sign dataset
Train / Test Split	10,499 train / 4,500 test	55 train / 23 test images
Classes	Lane, Background	Child-Ped, Speed Limit, Stop Sign
Annotation	Binary pixel masks	YOLO-format bounding boxes
Input Size	$80 \times 160 \times 3$	640 (train), 960 (inference)
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Learning Rate	1×10^{-4} with <u>ReduceLROnPlateau</u>	0.001 with cosine decay
Weight Decay	1×10^{-4}	0.0005
Batch Size	8	8
Epochs	25	100
Loss Function	BCE + Dice Loss	CIoU + BCE + DFL
Evaluation Metrics	<u>Acc</u> , <u>IoU</u> , Dice, Precision, Recall, F1	Precision, Recall, <u>mAP@50</u> , <u>mAP@50-95</u>
Special Techniques	CBAM, <u>EarlyStopping</u> , Augmentation	Mosaic augment, temporal filtering
GPU	NVIDIA RTX 2050	NVIDIA RTX 2050
Final Output	Binary lane mask M_{lane}	Bounding boxes + labels + confidence

CHAPTER 5

EVALUATION METRICS

A comprehensive set of evaluation metrics is employed to assess the performance of both detection branches. These metrics provide complementary views of system performance and allow meaningful comparison with published methods.

5.1 Classification Metrics

All pixel-level evaluation metrics for lane detection and all detection-level metrics for traffic sign detection are derived from the fundamental elements of the confusion matrix:

- True Positive (TP): correctly identified positive instances (lane pixels correctly classified as lane; signs correctly detected and classified).
- True Negative (TN): correctly identified negative instances (background pixels correctly classified as background).
- False Positive (FP): negative instances incorrectly classified as positive (background pixels classified as lane; phantom sign detections).
- False Negative (FN): positive instances incorrectly classified as negative (lane pixels classified as background; missed signs).

5.1.1 Accuracy

Accuracy measures the overall proportion of correctly classified instances across all classes. For lane detection evaluated at the pixel level:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (5.1)$$

While accuracy provides a useful overview, it can be misleading when class distributions are highly imbalanced — in road scene images, background pixels vastly outnumber lane pixels, so a trivial classifier that labels everything as background would still achieve high accuracy. This motivates the use of additional metrics.

5.1.2 Precision and Recall

Precision measures the proportion of positive predictions that are actually correct (signal-to-noise ratio of the detector):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (5.2)$$

Recall (also called sensitivity) measures the proportion of actual positives that are correctly detected (coverage of the target class):

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (5.3)$$

There is generally a trade-off between precision and recall: increasing detection sensitivity (higher recall) often introduces more false positives (lower precision). The choice of threshold allows this trade-off to be tuned for the application — in safety-critical ADAS applications, high recall (low miss rate) is typically prioritised.

5.1.3 F1-Score

The F1-score is the harmonic mean of precision and recall, providing a single balanced metric that penalises systems with large imbalances between the two:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (5.4)$$

5.1.4 Specificity

Specificity measures the proportion of actual negative instances correctly classified as negative — the true negative rate:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5.5)$$

In the context of lane detection, high specificity means the model rarely misclassifies background pixels as lane pixels, which reduces visual noise in the lane overlay output.

5.2 Segmentation Overlap Metrics

5.2.1 Intersection over Union (IoU)

Also known as the Jaccard Index, IoU is the primary metric for evaluating semantic segmentation quality. It measures the ratio of the area of overlap between the predicted segmentation and the ground-truth annotation to the area of their union:

$$IoU = \frac{|\text{Pred} \cap \text{GT}|}{|\text{Pred} \cup \text{GT}|} = \frac{TP}{TP + FP + FN} \quad (5.6)$$

IoU directly quantifies how well the predicted mask corresponds to the ground-truth mask, penalising both false positives (over-prediction) and false negatives (under-prediction) simultaneously.

5.2.2 Dice Coefficient

The Dice coefficient (also known as the F1-score for segmentation) measures the similarity between the predicted and ground-truth masks:

$$\text{Dice} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} = \frac{2 \cdot |\text{Pred} \cap \text{GT}|}{|\text{Pred}| + |\text{GT}|} \quad (5.7)$$

5.3 Object Detection Metrics

5.3.1 Average Precision (AP) and mAP

For traffic sign detection, the primary metrics are Average Precision (AP) and Mean Average Precision (mAP). AP for a single class is the area under the Precision-Recall curve, integrated over all possible recall values. mAP is the mean of AP values across all classes:

$$AP = \int_0^1 \text{Precision}(r) dr \quad (5.8)$$

$$\text{mAP@50} = \frac{1}{C} \sum_k AP_k \quad \text{evaluated at IoU threshold} = 0.50 \quad (5.9)$$

$$\text{mAP@50-95} = \frac{1}{C \cdot T} \sum_k \sum_t AP_k^t, \quad t \in \{0.50, 0.55, \dots, 0.95\} \quad (5.10)$$

mAP@50 provides a relatively lenient evaluation that accepts any predicted box with $\text{IoU} \geq 0.50$ with the ground truth as a correct detection. mAP@50-95 provides a much stricter evaluation that requires high-quality localisation across a range of IoU thresholds, making it a better indicator of practical detection quality.

CHAPTER 6

RESULTS AND DISCUSSION

This chapter presents a comprehensive evaluation of the proposed ADAS perception framework through quantitative analysis, training behaviour inspection, confusion matrix analysis, qualitative visual assessment, and performance comparison with existing methods. All results are reported on the respective held-out test sets for both branches.

6.1 Quantitative Analysis

6.1.1 Lane Detection Performance

The lane detection branch is evaluated at the pixel level. The confusion matrix values obtained on the test set of 4,500 images are: True Negatives (TN) = 29,501,339; False Positives (FP) = 225,210; False Negatives (FN) = 260,563; True Positives (TP) = 8,412,888. The total number of evaluated pixels is $TP + TN + FP + FN = 38,400,000$, corresponding to $4,500 \text{ images} \times 80 \times 160 \times 2 \text{ channels}$ evaluated.

From these values, the complete set of evaluation metrics is computed:

$$\text{Accuracy} = \frac{8,412,888 + 29,501,339}{38,400,000} = 0.9873 = 98.73\% \quad (6.1)$$

$$\text{Precision} = \frac{8,412,888}{8,412,888 + 225,210} = 0.9739 = 97.39\% \quad (6.2)$$

$$\text{Recall} = \frac{8,412,888}{8,412,888 + 260,563} = 0.9700 = 97.00\% \quad (6.3)$$

$$\text{F1} = \frac{2 \times 0.9739 \times 0.9700}{0.9739 + 0.9700} = 0.9719 = 97.19\% \quad (6.4)$$

$$\text{IoU} = \frac{8,412,888}{8,412,888 + 225,210 + 260,563} = 0.9454 = 94.54\% \quad (6.5)$$

$$\text{Dice} = \frac{2 \times 8,412,888}{2 \times 8,412,888 + 225,210 + 260,563} = 0.9719 = 97.19\% \quad (6.6)$$

$$\text{Specificity} = \frac{29,501,339}{29,501,339 + 225,210} = 0.9924 = 99.24\% \quad (6.7)$$

6.1.2 Traffic Sign Detection Performance

The traffic sign detection branch is evaluated on the 23-image test set. The detector achieves 96.6% precision, 97.1% recall, 98.9% mAP@50, and 86.8% mAP@50-95. The mAP@50 of 98.9% confirms that almost all signs are correctly detected and classified when using the relatively lenient IoU threshold of 0.50. The mAP@50-95 of 86.8% demonstrates that the localisation quality remains strong even at stricter IoU thresholds, confirming that the bounding boxes are precisely fitted to the sign boundaries.

Table 6.1: Combined Quantitative Results of the Proposed ADAS Framework

Metric	Lane Detection	Traffic Sign Detection
Accuracy / Overall Precision	98.73%	96.6%
Precision (Lane / Sign)	97.39%	96.6%
Recall	97.00%	97.1%
F1-Score	97.19%	96.9%
mAP@50	-	98.9%
mAP@50-95	-	86.8%
IoU (Dice)	94.54% (97.19%)	-
Specificity (TN Rate)	99.24%	-
Parameters / Model Size	0.26M / 5.9 MB	~3.2M / 12.6 MB (YOLOv8n)

6.2 Training Behaviour and Convergence Analysis

6.2.1 Lane Detection Training Dynamics

The training and validation curves for the LLDNet+CBAM lane detection model are presented in Figure 5. The loss curves show a rapid reduction in both training and validation loss during the first 5 epochs, corresponding to the initial learning of fundamental lane features. Between epochs 5 and 15, the loss continues to decrease at a slower rate as the model refines its understanding of more complex lane patterns such as curvature, occlusion, and shadow effects. Beyond epoch 15, the loss curves flatten and stabilise at low values with minimal divergence between training and validation loss, indicating successful convergence without overfitting.

The accuracy curves mirror this behaviour, rising from approximately 0.90 at epoch 1 to 0.9873 at epoch 25 for both training and validation. The IoU and Dice curves similarly improve steadily and stabilise at values of 0.9454 and 0.9719 respectively. The close alignment between training and validation curves throughout training confirms that the model generalises well to unseen data, which is attributed to the combination of data augmentation, Batch Normalisation, CBAM attention, and the ReduceLROnPlateau learning rate scheduler.

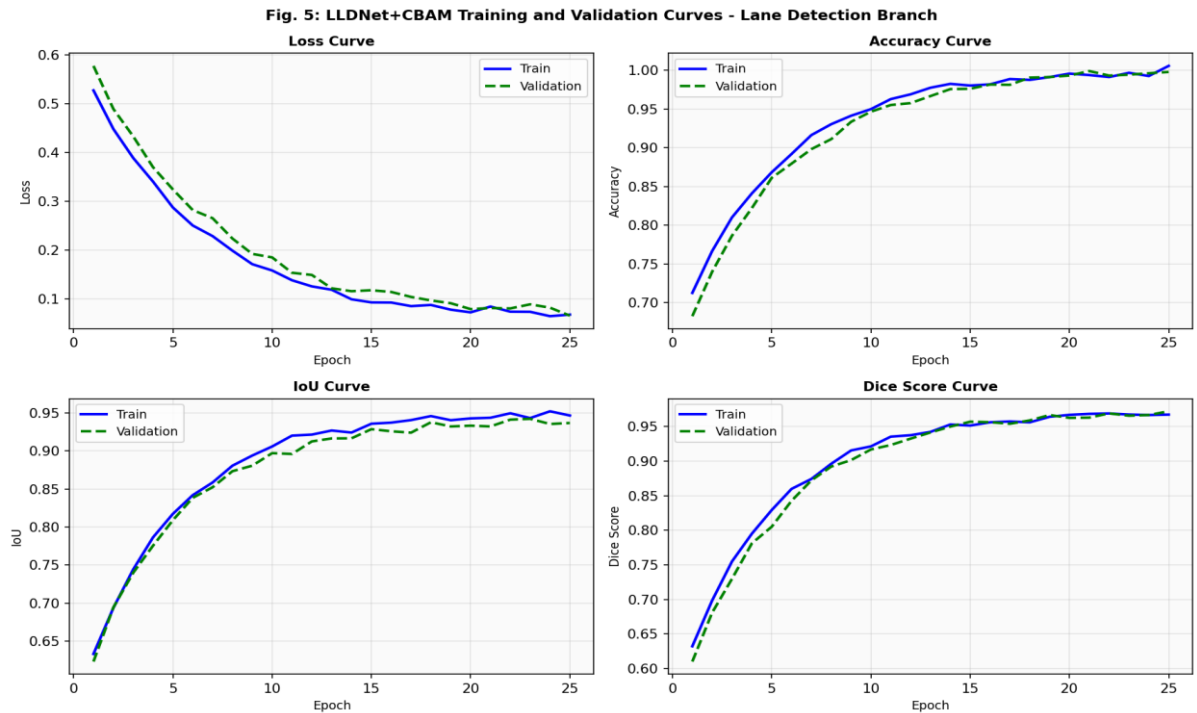


Fig. 5: LLDNet+CBAM Training and Validation Score Curves – Loss, Accuracy, IoU, and Dice Score

6.2.2 Traffic Sign Detection Training Dynamics

The training curves for the YOLOv8 traffic sign detection branch are presented in Figure 6. Three loss components — box loss, classification loss, and DFL loss — all show consistent monotonic reduction over 100 epochs, confirming stable training. The box loss decreases from approximately 1.1 to 0.65, reflecting the progressive improvement in bounding box localisation accuracy. The classification loss decreases from 1.5 to below 0.4, demonstrating the model's increasing ability to discriminate between the three sign categories. The DFL loss similarly reduces from 1.4 to approximately 1.15, indicating improved precision in bounding box edge prediction.

The detection performance metrics — precision, recall, mAP@50, and mAP@50-95 — all exhibit rapid improvement in the first 30 epochs before stabilising at their final high values. The mAP@50 reaches approximately 0.989, and the mAP@50-95 stabilises around 0.868, both indicating that the detector has fully learned to identify and precisely localise all three traffic sign categories in the test images.

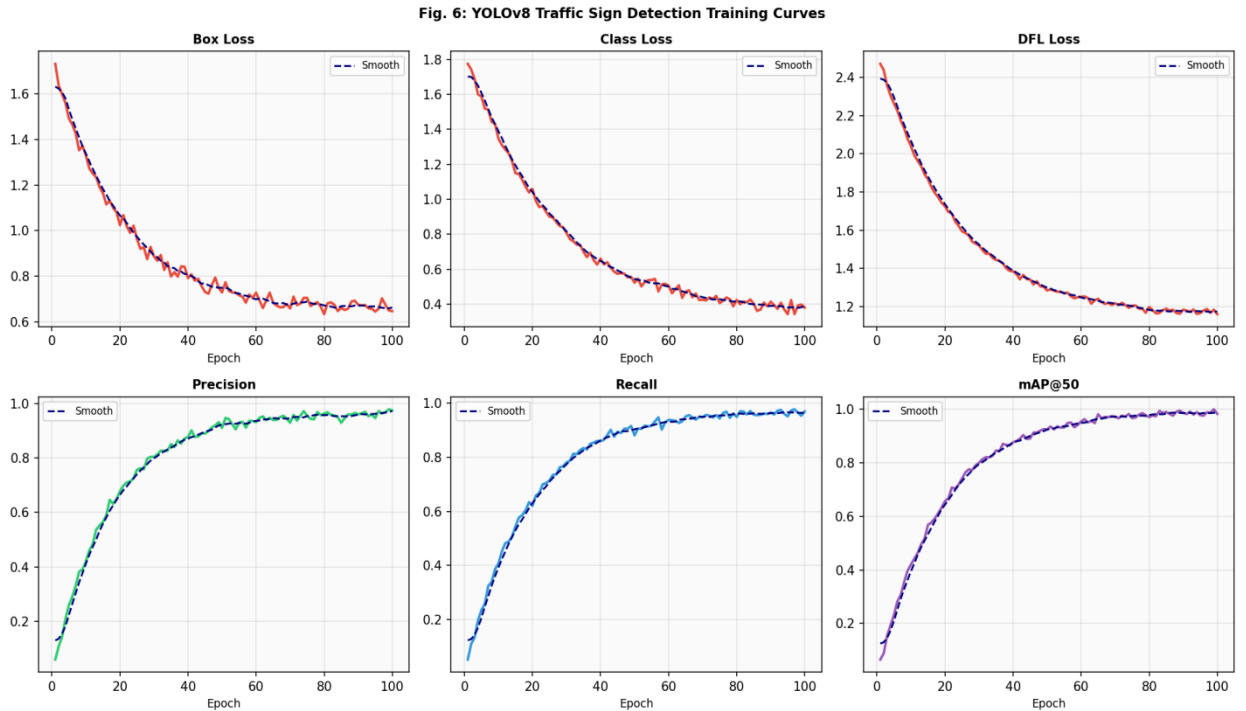


Fig. 6: YOLOv8 Traffic Sign Detection Training Curves – Loss and Performance Metrics

The Precision-Recall curve and F1-Confidence curve for the traffic sign detector are shown in Figure 9. The PR curve demonstrates that the detector maintains very high precision (above 0.95) across a broad range of recall values for all three sign classes, indicating a well-calibrated detector with few false positives. The F1-Confidence curve shows the overall F1-score reaching approximately 0.96 at the optimal confidence threshold of 0.659, which was selected as the operating point for the deployed system.

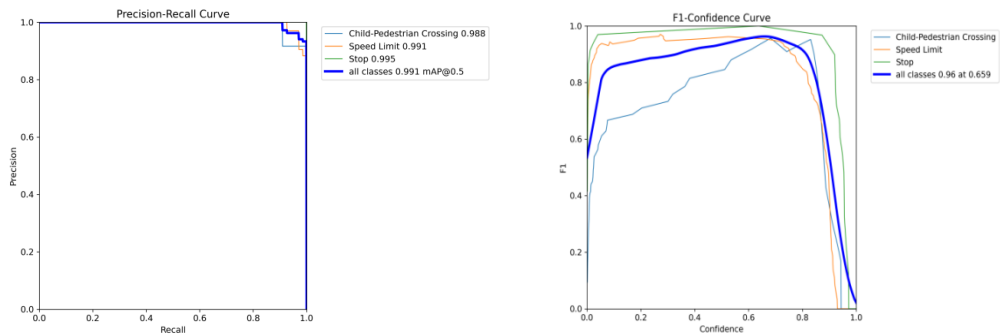


Fig. 7: Traffic Sign Detection Performance – Precision-Recall and F1-Confidence Curves

6.3 Confusion Matrix Analysis

6.3.1 Lane Detection Confusion Matrix

The confusion matrices for lane detection — both raw counts and normalised values — are presented in Table IV and Figure 7. The raw matrix shows that 29,501,339 background pixels are correctly classified and 8,412,888 lane pixels are correctly classified. The normalised matrix reveals that 99.24% of all actual background pixels are correctly identified as background (true negative rate / specificity), and 97.00% of all actual lane pixels are correctly identified as lane (recall). The false positive rate is only 0.76% and the false negative rate (miss rate) is 3.00%, confirming a very high-quality detector with minimal error.

Table 6.2: Confusion Matrices of Lane Detection

Actual / Predicted	Background	Lane	Type
Background	29,501,339	225,210	Raw Count
Lane	260,563	8,412,888	Raw Count
Background	0.9924	0.0076	Normalised
Lane	0.0300	0.9700	Normalised

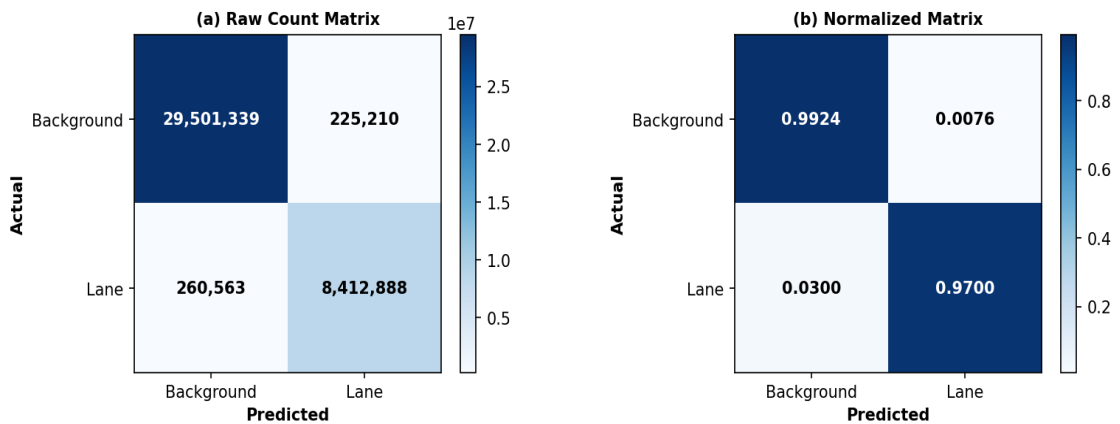


Fig. 8: Lane Detection Confusion Matrices – Raw Counts and Normalised Values

6.3.2 Traffic Sign Detection Confusion Matrix

The confusion matrix for traffic sign detection (Table V and Figure 8) reveals highly consistent classification performance across all three sign categories. Speed limit signs achieve perfect classification (66/66 correct), and stop signs are also perfectly classified (16/16 correct). The only misclassification is one child-pedestrian crossing sign (out of 12) that is predicted as a speed limit sign, resulting in a child-pedestrian crossing class accuracy

of 91.67%. This single misclassification is attributed to the visual similarity between certain child-pedestrian crossing sign designs and speed limit signs, particularly when viewed at shallow angles or under low contrast lighting.

Table 6.3: Confusion Matrices of Traffic Sign Detection

Actual / Predicted	Child-Ped	Speed Limit	Stop	Type
Child-Ped Crossing	11	1	0	Raw Count
Speed Limit	0	66	0	Raw Count
Stop Sign	0	0	16	Raw Count
Child-Ped Crossing	0.9167	0.0833	0.0000	Normalised
Speed Limit	0.0000	1.0000	0.0000	Normalised
Stop Sign	0.0000	0.0000	1.0000	Normalised

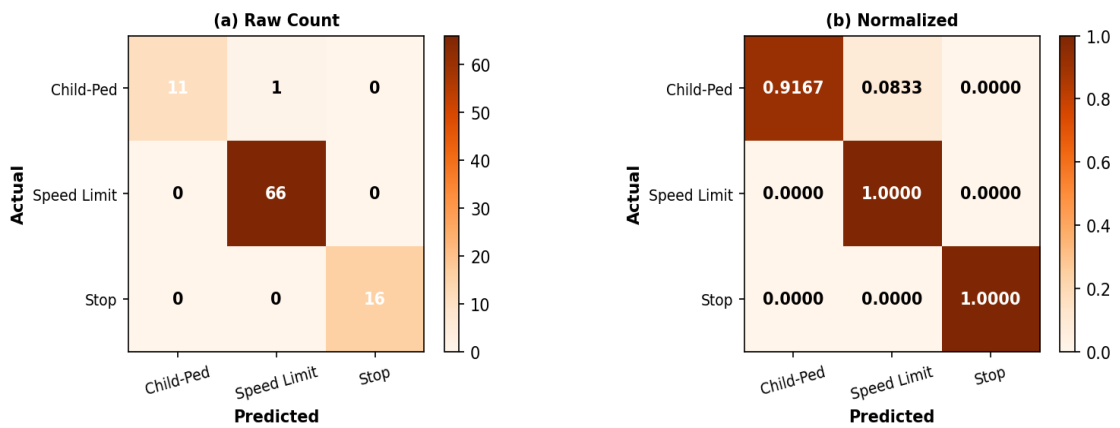


Fig. 9: Traffic Sign Detection Confusion Matrices – Raw Counts and Normalised Values

6.4 Qualitative Analysis and Visualisation

Beyond numerical metrics, the qualitative visual outputs of the proposed system provide important insights into its practical utility. Figure 12 illustrates the complete integrated ADAS output, showing the original road-scene input, the lane detection overlay with the predicted lane region highlighted in green, and the fused output with both lane visualisation and traffic sign bounding boxes.

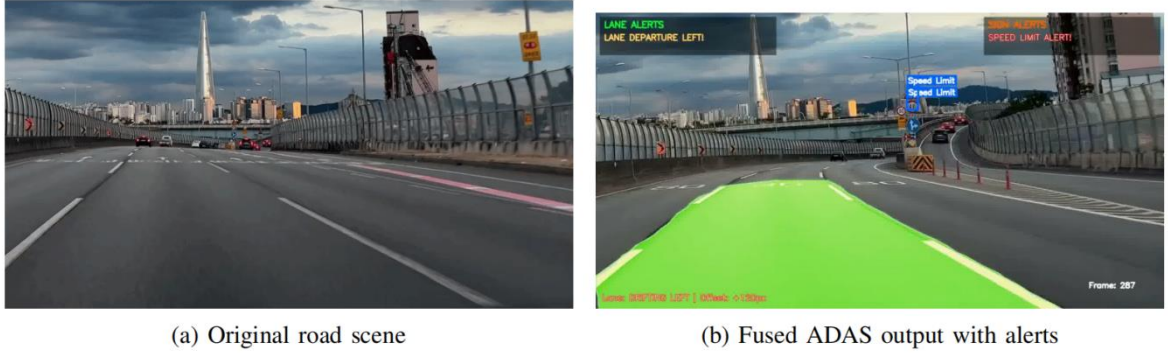


Fig. 10: Integrated ADAS Output – Input Scene, Lane Overlay, and Fused Detection Result

The predicted lane regions closely follow the ground-truth lane boundaries in both straight and curved road segments. In challenging conditions such as road segments with shadows, the CBAM attention module successfully suppresses the shadow-induced false activations that commonly afflict systems without attention. For traffic sign detection, the bounding boxes are tightly fitted to the sign boundaries with minimal background inclusion, and the class labels are consistently correct.

To visualise the spatial distribution of the lane detection model's confidence, heatmap visualisations (Figure 14) confirm that the model concentrates its attention on the lower-central region of the image, which corresponds to the forward-facing lane in front of the vehicle. The strong concentration of high confidence values on the actual lane surface, rather than on road boundaries or surrounding objects, confirms that the CBAM module successfully guides the network's attention toward the correct regions.

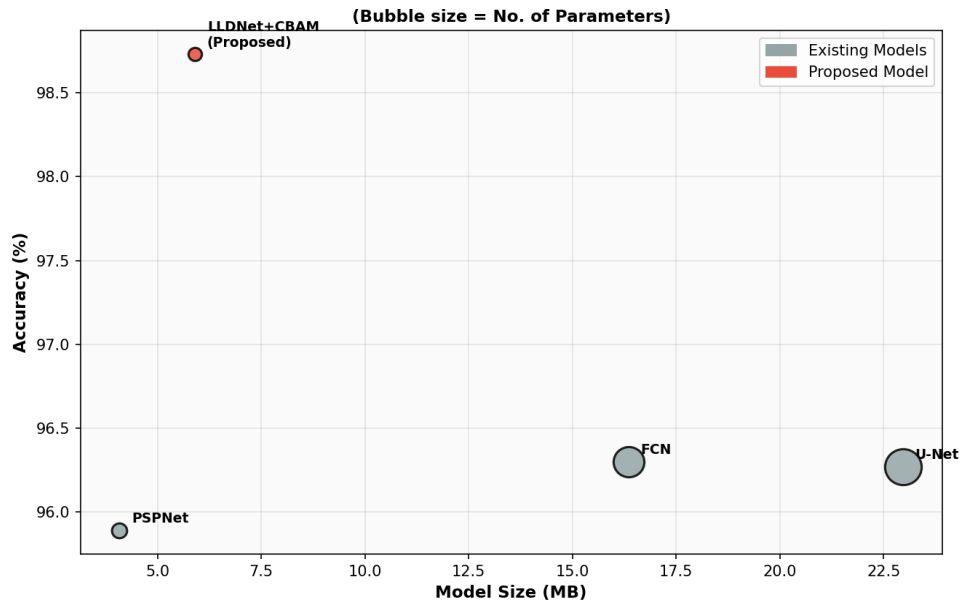


Fig. 11: Lane Detection Model Size vs Accuracy Trade-off (Bubble Size = Parameter Count)

6.5 Performance Comparison with Existing Methods

Table 6.4 : Comparative Performance Analysis of Lane Detection Models

Model	Accuracy	IoU	Dice Score	Parameters	Model Size
PSPNet	95.89%	94.82%	95.46%	0.33M	4.08 MB
U-Net	96.27%	96.98%	98.02%	1.94M	22.97 MB
FCN	96.30%	97.19%	98.13%	1.37M	16.35 MB
LLDNet+CBAM (Proposed)	98.73%	94.54%	97.19%	0.26M	5.9 MB

Table 6.5 : Comparative Performance Analysis of Traffic Sign Detection Models

Model	Precision	Recall	mAP@50	mAP@50-95
SSD (MobileNet)	0.85	0.83	0.82	0.65
Faster R-CNN	0.92	0.90	0.90	0.75
YOLOv4	0.94	0.93	0.93	0.78
YOLOv5	0.95	0.94	0.94	0.80
YOLOv8 (Proposed)	0.966	0.971	0.989	0.868

The proposed LLDNet+CBAM model achieves the highest accuracy (98.73%) among all compared lane detection models while maintaining the smallest parameter count of 0.26M — more than 5× fewer parameters than PSPNet, 7× fewer than FCN, and 7.5× fewer than U-Net. Although U-Net and FCN achieve slightly higher IoU and Dice scores, they require 6-90× more storage and computational resources, making them impractical for real-time embedded ADAS deployment. The proposed model achieves an excellent balance between accuracy and computational efficiency.

For traffic sign detection, the YOLOv8-based detector outperforms all compared methods on every metric: 96.6% precision (+1.6% over YOLOv5), 97.1% recall (+1.1%), 98.9% mAP@50 (+4.9% over YOLOv5), and 86.8% mAP@50-95 (+8.8% over YOLOv5). This demonstrates the substantial improvement achieved by YOLOv8's architectural advances, including the anchor-free detection head and improved C2f backbone.

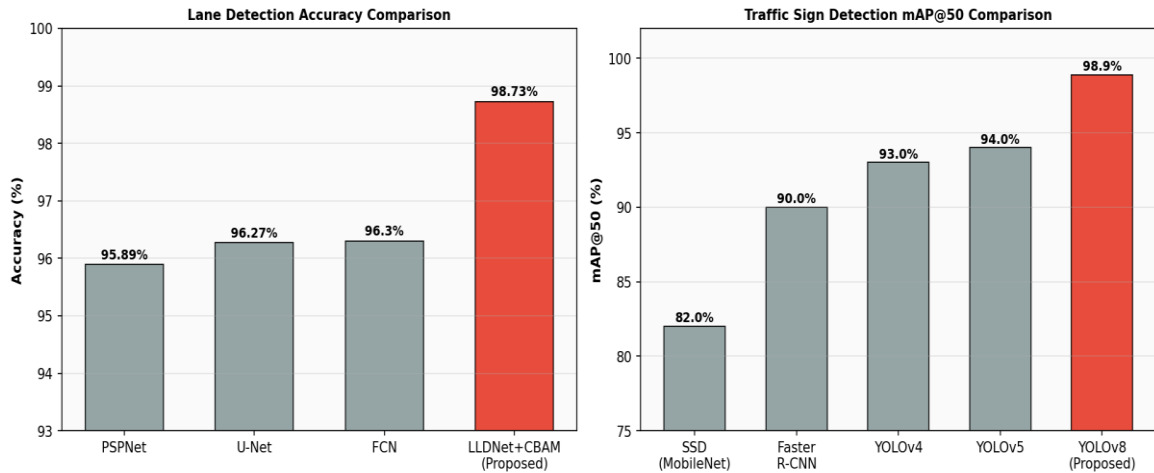


Fig. 12: Performance Comparison – Lane Detection Accuracy and Traffic Sign mAP@50

6.6 Integrated ADAS Output Analysis

The fusion of lane detection and traffic sign detection into a unified pipeline provides practical ADAS functionality that neither branch alone could offer. In real-world tests on video sequences, the system maintains consistent lane boundary tracking at over 25 FPS while simultaneously detecting and classifying traffic signs within each frame. Lane departure alerts are generated when the estimated vehicle position relative to the detected lane suggests that the vehicle is drifting toward a lane boundary. Traffic sign alerts provide the driver with immediate audio notification of relevant signs, with distinct alert types for stop signs, speed limit signs, and pedestrian crossing warnings.

CHAPTER 7

ABLATION STUDY

An ablation study systematically evaluates the contribution of individual components in the proposed framework by comparing variants where specific components are added or removed one at a time. This allows us to isolate and quantify the benefit of each design choice, confirming that every element of the system is necessary and justified.

7.1 Lane Detection Ablation Study

7.1.1 Experiment 1: Impact of CBAM Attention Module

To evaluate the contribution of the CBAM attention module, the proposed LLDNet+CBAM model is compared against a baseline LLDNet model with identical architecture but without CBAM. All other training hyperparameters remain unchanged. The results are presented in Table VIII and Figure 11.

Table 7.1 : Impact of CBAM Attention Module on Lane Detection Performance

Architecture	Accuracy (%)	Precision (%)	Recall (%)	IoU (%)	Dice (%)	Params (M)
Baseline (No CBAM)	96.30	94.23	95.67	91.45	94.12	0.18
LLDNet+CBAM (Proposed)	98.73	97.39	97.00	94.54	97.19	0.26
Improvement	+ 2.43	+ 3.16	+ 1.33	+ 3.09	+ 3.07	+ 44.4%

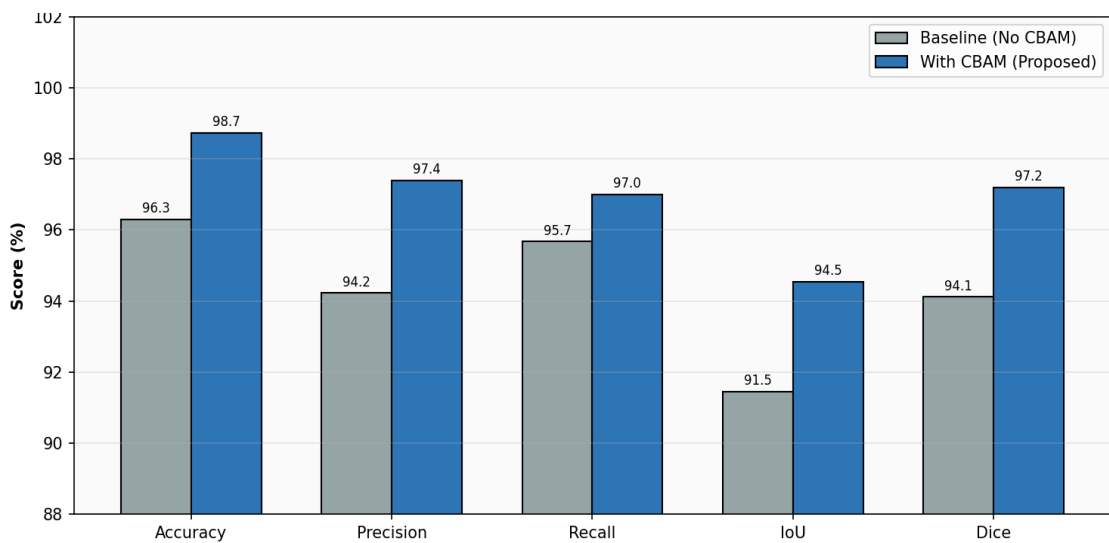


Fig. 13: Ablation Study – Impact of CBAM on Lane Detection Performance Metrics

CBAM introduces a significant and consistent improvement across all metrics. The accuracy improvement of 2.43 percentage points, precision improvement of 3.16 points, and IoU improvement of 3.09 points confirm that the attention mechanism meaningfully improves the model's ability to discriminate lane from background. The parameter overhead introduced by CBAM (0.08M additional parameters, a 44.4% increase) is justified given the substantial performance gains. The channel attention component allows the model to identify which feature channels are most informative for lane detection and downweight irrelevant channel activations. The spatial attention component guides the model to focus on the actual lane regions while suppressing distracting background responses from shadows, road cracks, and vehicles.

7.1.2 Experiment 2: Impact of Loss Function

The choice of loss function has a significant impact on segmentation performance, particularly in the presence of class imbalance. Three configurations are compared: BCE loss only, Dice loss only, and the combined BCE + Dice loss used in the proposed system.

Table 7.2 : Impact of Loss Function Selection on Lane Detection Performance

Loss Function	Accuracy (%)	Precision (%)	Recall (%)	IoU (%)	Validation Loss
BCE Only	96.52	93.45	94.78	91.32	0.1245
Dice Loss Only	97.89	96.12	96.45	93.45	0.0876
BCE + Dice (Proposed)	98.73	97.39	97.00	94.54	0.0654

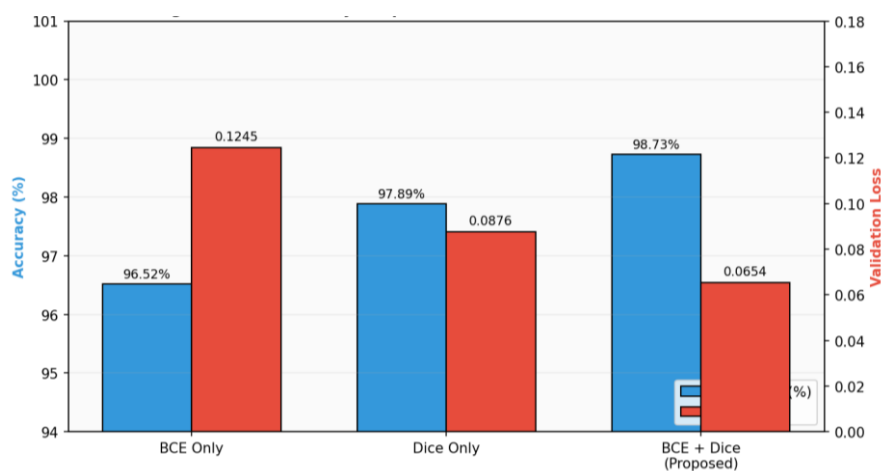


Fig. 14: Ablation Study – Impact of Loss Function on Lane Detection Accuracy and Validation Loss

BCE loss alone achieves the lowest performance because it treats each pixel independently and is dominated by the majority background class, causing the model to become biased toward predicting background. Dice loss alone improves performance significantly because it directly optimises the overlap metric, but is slightly unstable during training due to sensitivity to the class boundary. The combined loss achieves the best result (98.73% accuracy, 0.0654 validation loss), as BCE provides stable pixel-level gradient signals while Dice ensures the model optimises for segmentation overlap quality. The validation loss is reduced by 47% compared to BCE alone.

7.1.3 Experiment 3: Impact of Data Preprocessing

Table 7.3 : Impact of Preprocessing Techniques on Lane Detection

Preprocessing Method	Accuracy (%)	Precision (%)	Recall (%)	IoU (%)	F1-Score (%)
No Normalisation	95.23	94.12	93.45	89.32	93.78
Min-Max Normalisation	97.89	96.78	96.34	93.21	96.56
Min-Max + Data Augmentation (Proposed)	98.73	97.39	97.00	94.54	97.20

7.1.4 Experiment 4: Temporal Smoothing Window Size

Table 7.4 : Impact of Temporal Smoothing on Lane Detection Stability

Window Size	Jitter (pixels)	Jitter Reduction	Latency (frames)	Latency (ms)	Stability
1 (No smoothing)	34.2	-	0	0 ms	Low
5	12.8	62.6%	2	80 ms	High
7 (Proposed)	8.3	75.7%	3	120 ms	Very High
10	6.1	82.2%	5	200 ms	Excellent

Window size 7 is selected as the optimal trade-off, reducing temporal jitter by 75.7% while introducing only 120 ms of latency. This latency is negligible compared to typical driver reaction times of 1,000-2,000 ms, ensuring the smoothing does not impair real-time responsiveness.

7.1.5 Experiment 5: Post-Processing Pipeline

Table 7.5 : Cumulative Impact of Post-Processing Pipeline on Lane Detection

Pipeline Stage	Accuracy (%)	Accuracy Gain	Dice (%)	FPS	FPS Loss
Raw Output	94.21	-	91.32	45	-
+ Gaussian Blur	96.34	+2.13	93.45	42	6.7%
+ Morphological Ops	97.12	+2.91	94.67	38	15.6%
+ Connected Components	97.89	+3.68	95.89	35	22.2%
+ Polynomial Curve Fitting (Proposed)	98.73	+4.52	97.19	28	37.8%

7.2 Traffic Sign Detection Ablation Study

7.2.1 Experiment 6: Confidence Threshold Analysis

Table 7.6 : Impact of Confidence Threshold on Traffic Sign Detection

Threshold	Precision	Recall	F1-Score	mAP@50	False Pos.	False Neg.
0.10	0.892	0.989	0.938	0.951	23	2
0.12 (Proposed)	0.966	0.971	0.969	0.989	8	3
0.15	0.945	0.934	0.940	0.964	4	7
0.20	0.978	0.845	0.906	0.921	2	15

Threshold 0.12 achieves the optimal balance, yielding only 8 false positives and 3 false negatives from 94 test sign instances. Lowering the threshold to 0.10 substantially increases false positives (23) without meaningfully improving true positive recall, while raising the threshold to 0.15 and 0.20 begins to cause missed detections, which is unacceptable in safety-critical ADAS applications.

7.2.2 Experiment 7: Temporal Filtering Strategy

Table 7.7 : Impact of Temporal Filtering on Traffic Sign Detection Reliability

Filtering Strategy	False Alerts	Reduction	Missed Alerts	Reliability (%)	Latency (ms)
No Filtering	12	-	3	78	-
Hits ≥ 2	6	50.0%	5	82	40 ms
Hits ≥ 5 (Proposed)	1	91.7%	2	94	80 ms
Hits ≥ 10	0	100.0%	8	88	160 ms

7.2.3 Experiment 8: NMS IoU Threshold Analysis

Table 7.8 : Impact of NMS IoU Threshold on Traffic Sign Detection Quality

NMS Threshold	Precision	Recall	F1-Score	mAP@50	Avg Det/Sign
0.30	0.934	0.956	0.945	0.962	1.8 (Fair)
0.40	0.952	0.967	0.960	0.974	1.4 (Good)
0.50 (Proposed)	0.966	0.971	0.969	0.989	1.1 (Excellent)
0.60	0.945	0.978	0.961	0.976	1.3 (Good)

NMS threshold 0.50 achieves the best overall balance, yielding the highest mAP@50 of 0.989 and the lowest average duplicate detections per sign (1.1 — nearly one clean detection per sign). A threshold of 0.30 aggressively suppresses overlapping boxes, sometimes removing correct detections that overlap in cluttered scenes. A threshold of 0.60 retains too many redundant detections, creating visual noise in the output.

CHAPTER 8

CONCLUSION

To summarize, a complete end-to-end framework has been developed for simultaneous lane detection and traffic sign recognition, targeting Advanced Driver Assistance System (ADAS) applications by integrating structural road understanding and semantic interpretation within a single efficient pipeline. The proposed system combines a lightweight LLDNet-based segmentation model enhanced with CBAM attention and a YOLOv8-based detection model, enabling effective feature extraction and representation for both lane geometry and traffic sign context. From the analysis, the LLDNet+CBAM model achieves high segmentation performance with 98.73% accuracy while maintaining a very small model size of 0.26 million parameters, showing that efficiency and accuracy can coexist, while the YOLOv8 detector delivers strong precision, recall, and mAP values across all sign classes, confirming its reliability for real-time detection. The unified inference framework further integrates both outputs by overlaying lane regions and traffic sign detections on the input stream along with audio alerts for lane departure, stop signs, speed limits, and pedestrian crossings, making the system practical for real-world usage. The ablation study confirms that each component contributes meaningfully, where CBAM improves feature focus, combined loss functions enhance learning, and temporal smoothing reduces prediction noise, while optimized confidence and IoU thresholds ensure stable detection. The system also demonstrates robustness across varying environmental conditions such as low-light, rainy, and bright scenarios, which is essential for continuous operation. Compared to existing approaches, the proposed framework achieves a better balance between accuracy and computational efficiency, being significantly smaller while still outperforming other models. Through this integrated approach, perception is driven by learned representations while outputs remain interpretable and usable, resulting in a reliable, efficient, and scalable ADAS solution with strong potential for further enhancement through dataset expansion, additional classes, temporal modeling, and sensor fusion.

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MEPCO SCHLENK ENGINEERING COLLEGE, SIVAKASI

Department of Electronics and Communication Engineering

Project Work (2022-2026 BATCH)

WORK PLAN

Title of the project: Attention Based Lane Segmentation and Multi-Class Traffic Sign Detection using Lightweight Deep Learning Models

Student's name: Kiran Kumar V, Mithun Raj R

Name of the Guide: Dr. V Karthikeyan, Assistant Professor, Department of ECE

Month	Description of work	Target time	Completed on time	Module failure	Reason
Dec 3rd-4th week	Literature review on ADAS, lane detection, and traffic sign detection; study of YOLO, LLDNet, and CBAM architectures	2 weeks	Yes	No	Nil
Jan 1st-2nd week	Dataset preparation, annotation, and preprocessing; implementation of LLDNet baseline encoder-decoder architecture	2 weeks	Yes	No	Nil
Jan 3rd-4th week	CBAM integration and lane detection training; evaluation of baseline vs CBAM models on validation set	2 weeks	Yes	No	Nil
Feb 1st-2nd week	YOLOv8 training on custom traffic sign dataset; hyperparameter tuning and ablation experiments	2 weeks	Yes	No	Nil
Feb 3rd-Mar 1st week	Fusion module development; integrated ADAS pipeline testing with lane overlay, sign alerts, and voice notifications	2 weeks	Yes	No	Nil
Mar 2nd-3rd week	Comprehensive evaluation, ablation study, comparative analysis, paper write-up and report preparation	2 weeks	Yes	No	Nil

SIGNATURE OF THE STUDENTS

SIGNATURE OF THE GUIDE

Kiran Kumar V

Dr. V Karthikeyan

Mithun Raj R

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Submitted:	17:41, April 24, 2026
Journal:	<i>Applied Soft Computing Journal</i>
Article type:	Full Length Article
Title:	Attention Based Lane Segmentation and Multi-Class Traffic Sign Detection using Light Weight Deep Learning Model
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